Forest and Wildlife Resources

The big idea

We share this planet with millions of other living beings. Forests and wildlife give us oxygen, food, medicine, timber and a stable climate. India is one of the world's richest countries in biological diversity, but much of it is under threat. This chapter explains why, and how governments and **local communities** are trying to conserve it.

Flora and fauna in India

India is one of the **mega-biodiversity countries** of the world, with nearly **8 per cent** of the total number of species in the world (estimated at about 1.6 million). At least **10 per cent** of India's recorded wild flora and **20 per cent** of its mammals are on the threatened list.

IUCN CATEGORIES OF SPECIES (WITH INDIAN EXAMPLES)

- **Normal species**: populations considered normal for survival: cattle, sal, pine, rodents.
- **Endangered species**: in danger of extinction: **blackbuck, crocodile, Indian wild ass, Indian rhino, lion-tailed macaque, sangai** (brow-antlered deer in Manipur).
- **Vulnerable species**: population has declined to levels from where it is likely to move into the endangered category: **blue sheep, Asiatic elephant, Gangetic dolphin**.
- **Rare species**: species with small populations that may move into endangered/vulnerable categories: **Himalayan brown bear, wild Asiatic buffalo, desert fox and hornbill**.

- **Endemic species**: found only in particular areas, usually isolated by natural or geographical barriers: **Andaman teal, Nicobar pigeon, Andaman wild pig, mithun** in Arunachal Pradesh.
- **Extinct species**: not found after searches of known or likely areas: **Asiatic cheetah, pink head duck**.

THE ASIATIC CHEETAH

The world's fastest land mammal was declared **extinct in India in 1952**. (Cheetahs from Africa were reintroduced in Kuno National Park, Madhya Pradesh, in 2022, but that is a different subspecies.)

Causes of depletion

- **Colonial period**: expansion of railways, agriculture, commercial and scientific forestry and mining.
- **Agricultural expansion**: between 1951 and 1980, over 26,200 sq km of forest area was converted to agricultural land.
- **Shifting cultivation** (slash and burn) in tribal belts of the north-east and central India.
- **Large-scale development projects**: since 1951, over **5,000 sq km** of forest was cleared for river valley projects; the **Narmada Sagar Project** in Madhya Pradesh would inundate about **40,000 hectares** of forest.
- **Mining**: for example, the **Buxa Tiger Reserve** in West Bengal is seriously threatened by ongoing **dolomite mining**, disturbing the natural habitat of many species and blocking the migration route of the great Indian elephant.
- **Enrichment plantations**: large areas of natural forest replaced by single commercially valuable species (teak in south India; chir pine in the Himalayas).
- Grazing, fuelwood collection, poaching, pollution and forest fires, and the unequal consumption of resources (the richest 5% of society may cause more ecological damage than the poorest 25%).

THE HIMALAYAN YEW

The Himalayan yew (*Taxus wallachiana*), a medicinal plant found in Himachal Pradesh and Arunachal Pradesh, yields **taxol**, a chemical used as an anti-cancer drug. Over-exploitation has made it endangered; thousands of trees have dried up.

Loss of forests also has cultural consequences: **women** in poor households often have to walk long distances to collect fuel, fodder and water, increasing their workload and health problems.

Conservation of forests and wildlife

Conservation preserves ecological diversity and life-support systems: water, air and soil. It also preserves the genetic diversity of plants and animals for better growth of species and breeding.

KEY MEASURES BY THE GOVERNMENT

- **Indian Wildlife (Protection) Act, 1972**: an all-India list of protected species was published; hunting was banned; trade in wildlife was restricted; habitats were given legal protection; national parks and wildlife sanctuaries were set up.
- **Project Tiger (1973)**: launched because tiger numbers had dropped to about 1,827 in 1973. Threats: poaching for trade in skins and bones, shrinking habitat, depletion of prey. Examples of tiger reserves: **Corbett National Park (Uttarakhand), Sunderbans National Park (West Bengal), Bandhavgarh National Park (Madhya Pradesh), Sariska Wildlife Sanctuary (Rajasthan), Manas Tiger Reserve (Assam), Periyar Tiger Reserve (Kerala)**.
- Projects for other species, such as the Indian elephant, rhino, crocodile, and the Asiatic lion of Gir.

Types of forests (by government classification):

Type	What it is	States where it dominates
Reserved forests	More than **half** of total forest land; regarded as the most valuable for conservation	Jammu and Kashmir, Andhra Pradesh, Uttarakhand, Kerala, Tamil Nadu, West Bengal, Maharashtra
Protected forests	About **one-third** of forest area; protected from further depletion	Bihar, Haryana, Punjab, Himachal Pradesh, Odisha, Rajasthan
Unclassed forests	Other forests and wastelands belonging to government and private individuals and communities	North-eastern states and parts of Gujarat

Reserved and protected forests are called **permanent forest estates**. **Madhya Pradesh** has the largest area under permanent forests (75% of its total forest area).

Community and conservation

Local communities have often protected forests better than laws.

- **Sariska Tiger Reserve (Rajasthan)**: villagers fought against mining by citing the Wildlife Protection Act.
- **Alwar district (Rajasthan)**: inhabitants of five villages declared **1,200 hectares** of forest as the **Bhairodev Dakav 'Sonchuri'**, with their own rules banning hunting and protecting wildlife against outside encroachments.
- **Chipko Movement** in the Himalayas: villagers (many women) hugged trees to resist deforestation; it also showed community afforestation with indigenous species can succeed.
- **Beej Bachao Andolan** in **Tehri** and **Navdanya**: showed that adequate farm production is possible without synthetic chemicals, through diversified crops.

- **Joint Forest Management (JFM)**: a programme involving local communities in managing and restoring degraded forests. It was first passed as a resolution in **Odisha in 1988**. Communities get benefits such as non-timber forest produce and a share in timber harvested.

Sacred groves and traditions. Nature worship is an age-old tribal belief that all creations of nature have to be protected; such beliefs preserved virgin forests called **sacred groves** (forests of God and Goddesses). The **Mundas and Santhals** of Chota Nagpur worship **mahua** and **kadamba** trees; tribals of Odisha and Bihar worship **tamarind and mango** trees during weddings. The **peepal** and **banyan** are considered sacred by many. The **Bishnois** of Rajasthan protect **blackbuck (chinkara), nilgai and peacocks**, which can be seen in herds around their villages.

ENDANGERED VS VULNERABLE

The **Asiatic elephant** and **Gangetic dolphin** are **vulnerable**, not endangered; the blackbuck and Indian rhino are **endangered**. Mithun is **endemic** (Arunachal), not extinct.

BOARD FAVOURITES

"Classify species into IUCN categories with examples", "Describe the steps taken to conserve wildlife / Project Tiger", "Distinguish reserved and protected forests", "How have communities contributed to conservation?" (5 marks). In the current CBSE scheme this chapter may be assessed only internally, but its facts are often used in source-based questions.

Quick recap

- India: mega-biodiversity country with about 8% of world species.
- IUCN categories: normal, endangered, vulnerable, rare, endemic, extinct.
- Depletion: agriculture, development projects, mining (Buxa), shifting cultivation, grazing, poaching.

- Wildlife (Protection) Act 1972; Project Tiger 1973.
- Forests: reserved (> half), protected (about one-third), unclassed.
- Community efforts: Sariska, Bhairodev Dakav Sonchuri, Chipko, Beej Bachao, JFM (Odisha 1988), Bishnois.

Practice questions

1. Which of the following is an endemic species?
   (a) Blackbuck (b) Asiatic elephant (c) Andaman teal (d) Himalayan brown bear
2. Project Tiger was launched in
   (a) 1952 (b) 1972 (c) 1973 (d) 1988
3. The Buxa Tiger Reserve in West Bengal is threatened mainly by
   (a) dolomite mining (b) shifting cultivation (c) tea plantations (d) tsunami damage
4. Joint Forest Management was first passed as a resolution in
   (a) Rajasthan (b) Uttarakhand (c) Kerala (d) Odisha
5. The Bhairodev Dakav 'Sonchuri' was declared by villagers in
   (a) Tehri, Uttarakhand (b) Bastar, Chhattisgarh (c) Alwar district, Rajasthan (d) Kutch, Gujarat
6. The Gangetic dolphin falls under which IUCN category?
   (a) Endangered (b) Vulnerable (c) Rare (d) Extinct
7. Distinguish between reserved forests and protected forests.
8. "Local communities have played an important role in conserving forests and wildlife in India." Justify with examples.

Answers: 1. c 2. c 3. a 4. d 5. c 6. b 7. see notes 8. see notes